# NLP

25-06-26

I was learning NLP today

↓ tokenization

i was learn #ing NLP today

↓ token ids

$t_0$ $t_1$ $t_2$ $t_3$ $t_4$ $t_5$

↓ embedding layer

$e_0$ $e_1$ $e_2$ $e_3$ $e_4$ $e_5$ raw embeddings

\+ $p_0$ $p_1$ $p_2$ $p_3$ $p_4$ $p_5$ positional encodings

↓

$\bar{e}_0$ $\bar{e}_1$ $\bar{e}_2$ $\bar{e}_3$ $\bar{e}_4$ $\bar{e}_5$ position aware vectors

query matrix: $Q = \begin{bmatrix} \bar{e}_0 \\ \bar{e}_1 \\ \bar{e}_2 \\ \bar{e}_3 \\ \bar{e}_4 \\ \bar{e}_5 \end{bmatrix} W_Q$

$Q = \bar{e} W_Q$

$Q$: $6 \times d$ ; $\bar{e}$: $6 \times d$ ; $W_Q$: $d \times d$

Key matrix : $K = \bar{e} W_K$

$6 \times d \quad 6 \times d \quad d \times d$

Value Matrix : $V = \bar{e} W_V$

$6 \times d \quad 6 \times d \quad d \times d$

Dot products : $D = Q K^T$

$6 \times 6 \quad 6 \times d \quad d \times 6$

Attention scores : $\text{softmax}\left(\frac{D}{\sqrt{d}}\right) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)$

Context aware vectors : $\underbrace{\text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)}_{6 \times 6} \underbrace{V}_{6 \times d}$

$e' = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right) V$

$6 \times d$

i wad learn #ing NLP today

↓

self attention layer

↓

$e'_0 \quad e'_1 \quad e'_2 \quad e'_3 \quad e'_4 \quad e'_5$

## Cross Attention:

- used in encoder-decoder architecture

eg. Language translation

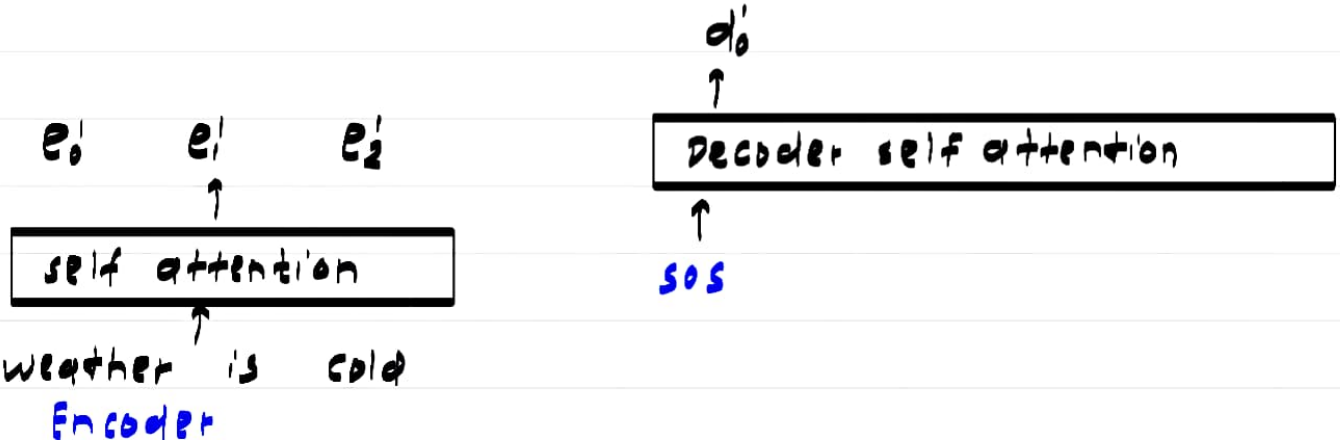
| Eng text | spanish text |
|---|---|
| weather is cold | el clima es frio |

input sequence (eng text) → [transformer model] → output sequence (spanish text)

## Steps in Encoder Decoder Transformers:

1) Apply self attention on input sequence and get their context aware vectors



every attention mechanism needs 3 entities

Q K V

query key value

in self attention : Q K V come from same sequence

in cross attention : K and V come from input sequence

Q comes from output sequence

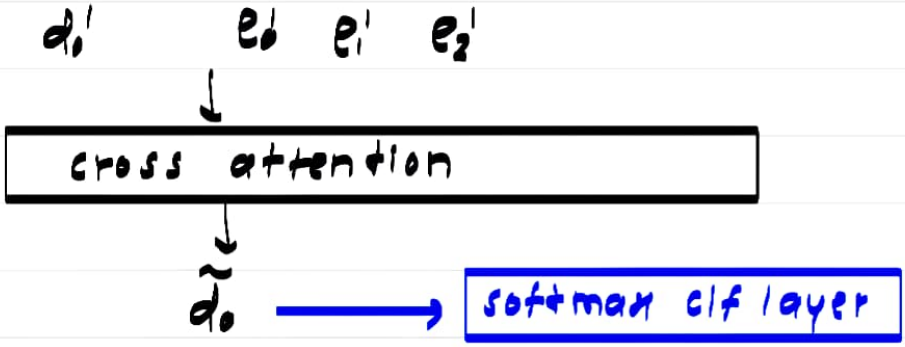
at t: 0

output seq = sos




input seq = weather is cold



inside cross attention:

$$Q = [d_0'] W_{CQ}$$

$1 \times d \quad 1 \times d \quad d \times d$

$$K = e' W_{CK} = \begin{bmatrix} e_0' \\ e_1' \\ e_2' \end{bmatrix} W_{CK}$$

$3 \times d \quad 3 \times d \quad d \times d$

$$V = e' W_{CV} = \begin{bmatrix} e_0' \\ e_1' \\ e_2' \end{bmatrix} W_{CV}$$

$3 \times d \quad 3 \times d \quad d \times d$

$e'$ → input sequence

$W_{CQ}$ $W_{CK}$ $W_{CV}$ → weights in cross attention layer

$$\tilde{d}_0 = \underbrace{\text{softmax}\left(\frac{Q K^T}{\sqrt{d}}\right)}_{1 \times 3} V$$

$1 \times d \quad 1 \times 3 \quad 3 \times d$

$$\tilde{d}_0 = \begin{matrix} \tilde{d}_{00} \\ \tilde{d}_{10} \\ \tilde{d}_{20} \\ \vdots \\ \tilde{d}_{d0} \end{matrix} \longrightarrow \begin{matrix} \bigcirc \\ \bigcirc \\ \bigcirc \\ \vdots \\ \text{20K nodes} \end{matrix} \xrightarrow{\text{softmax}} \begin{matrix} p_1 \\ p_2 \\ p_3 \\ p_4 \\ \vdots \\ p_{20K} \end{matrix}$$

20K tokens in french vocab

from $p_1$ to $p_{20K}$ → $p_{500}$ is the max probability

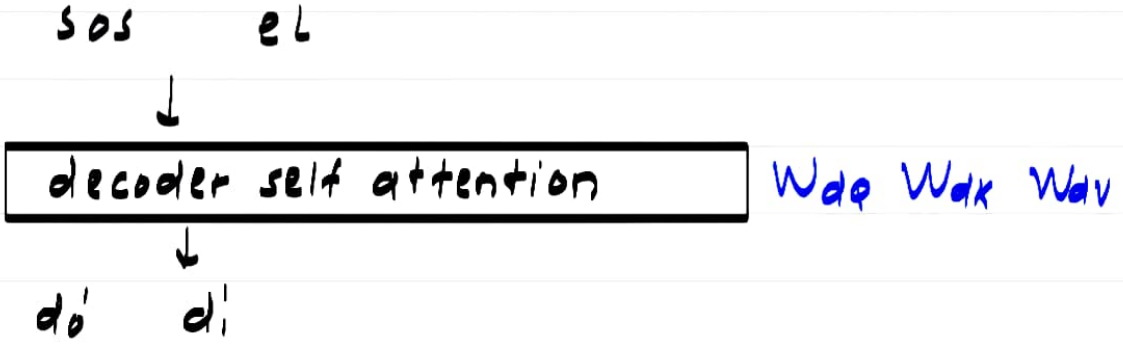
$p_{500} \rightarrow t_{500} =$ 'el'

token having max probability is generated by decoder

at t=7

we need Q K V for cross attention

in decoder self attention




$Q = [d_1'] W_{cq} \quad K = \bar{e} W_{ck} \quad V = \bar{e} W_{cv}$

$$\tilde{d}_1 = \text{softmax}\left(\frac{Q K^T}{\sqrt{d}}\right) V$$

$\tilde{d}_1 \rightarrow$ softmax clf layer $\rightarrow$ $p_1, p_2, p_3, \ldots, p_{20K}'$ } token with maximum probability → 'clima'

at $t = 2$

sos el clima

↓

decoder self attention

↓

$d_0'$ $d_1'$ $d_2'$

$Q = [d_2'] W_{cQ}$ $K = \bar{E} W_{cK}$ $V = \bar{E} W_{cV}$

$$\tilde{d}_2 = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right) V$$

$\tilde{d}_2 \rightarrow$ softmax clf layer $\rightarrow$ $p_1, p_2, p_3, \ldots, p_{20k}$ } token with max prob 'es'

at $t = 3$

sos el clima es

↓

decoder self attention

↓

$d_0'$ $d_1'$ $d_2'$ $d_3'$

$$\tilde{d}_3 = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right) V$$

$\tilde{d}_3 \rightarrow$ softmax clf layer $\rightarrow$ $p_1', p_2', \ldots, p_{20k}'$ } max probability token ↓ 'frío'

at $t = 4$

sos el clima es frio

↓

decoder self attention

↓

$d_0'$ $d_1'$ $d_2'$ $d_3'$ $d_4'$

$$\tilde{d_4} = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V$$

$\tilde{d_4}$ = softmax clf layer → $p_1, p_2, \ldots, p_{20K}$ } max prob token 'EOS'

EOS token stops the output sequence.

## Pre-training using transformer models:

more data + longer training + more weights

↓

really good LLM

LLM - Large Language model

I was eating an apple. It was great!

Masked language modeling

i <mask> eating an apple <sep> it was <mask>

↓ LLM

i was eating an apple <sep> it was great